Year 11 Foundation

# REVISION PACK

## Handling Data 4

**Name:** _____

**Class:** _____

## Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

**Worked example: frequency tables**

Here are the numbers of cars owned by 10 families. 2 0 1 1 3 1 2 1 0 2

(a) Complete a frequency table. (b) Write down the mode. (c) Work out the range. (d) Work out the mean.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Tally and count | $0:2,\ 1:4,\ 2:3,\ 3:1$ |
| **(b)** | **Step 1** | Most common | $1$ |
| **(c)** | **Step 1** | Biggest − smallest | $3-0=3$ |
| **(d)** | **Step 1** | Value × frequency, then add | $0+4+6+3=13$ |
| | **Step 2** | Divide by total frequency (10) | $13\div 10=1.3$ |

### Practice: frequency tables

**1.1**

| Number | Tally | Frequency |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

Here are the numbers of brothers and sisters of 20 students.

1 2 0 1 3 2 1 0 2 1

1 4 2 1 0 3 1 2 1 2

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of brothers and sisters.

**1.2**

| Number of books | Tally | Frequency |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

Here are the numbers of books read last month of 20 students.

2 3 1 0 2 2 4 1 3 2

0 1 2 3 2 1 2 4 3 2

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of books.

**1.3**

| Number of goals | Tally | Frequency |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

Here are the numbers of goals scored of 20 students.

0 1 1 2 0 3 1 1 0 2

1 4 1 0 2 1 3 1 2 1

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of goals.

## Practice: frequency tables (continued)

**1.4**

| Number of cousins | Tally | Frequency |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

Here are the numbers of cousins of 20 students.

2 1 3 0 2 4 1 2 3 2

1 0 2 3 1 2 4 2 1 3

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of cousins.

**1.5**

| Number of drinks | Tally | Frequency |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

Here are the numbers of drinks they had yesterday of 20 students.

1 0 2 1 1 3 2 1 0 1

2 1 4 1 0 2 1 3 1 2

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of drinks.

**1.6**

| Number of computers | Tally | Frequency |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

Here are the numbers of computers at home of 20 students.

0 2 1 1 0 3 1 2 0 1

1 0 2 1 4 1 0 2 1 1

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of computers.

## Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

**Worked example: pie charts from a table**

Information about 24 students is shown in a pie chart. 6 of the 24 students walk to school. Work out the angle of the sector for walking.

| | | |
|---|---|---|
| **Step 1** | Angle for one: 360° ÷ 24 | $360 \div 24 = 15^\circ$ |
| **Step 2** | × angle for one student | $6 \times 15 = 90^\circ$ |

### Practice: pie charts from a table

**1.7** Information about 30 students is shown in a pie chart. 7 of the 30 students cycle to school. Work out the angle of the sector for cycling.

**1.8** Information about 24 students is shown in a pie chart. 10 of the 24 students have one brother. Work out the angle of the sector for one brother.

**1.9** Information about 40 students is shown in a pie chart. 14 of the 40 students have two pets. Work out the angle of the sector for two pets.

**1.10** Information about 18 students is shown in a pie chart. 5 of the 18 students play tennis. Work out the angle of the sector for tennis.

**1.11** Information about 45 students is shown in a pie chart. 12 of the 45 students read every day. Work out the angle of the sector for reading every day.

**1.12** Information about 60 students is shown in a pie chart. 22 of the 60 students take the bus. Work out the angle of the sector for the bus.

## Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

**Worked example: the median of six numbers**

Find the median of these numbers. 4 9 1 7 6 3

| | | |
|---|---|---|
| **Step 1** | In order | $1, 3, 4, 6, 7, 9$ |
| **Step 2** | Two middle numbers | $4$ and $6$ |
| **Step 3** | Halfway | $(4 + 6) \div 2 = 5$ |

### Practice: the median of six numbers

**1.13** Find the median of these numbers. 5 2 8 4 10 7

**1.14** Find the median of these numbers. 12 3 9 6 15 4

**1.15** Find the median of these numbers. 7 1 5 9 3 6

**1.16** Find the median of these numbers. 10 14 8 12 6 11

**1.17** Find the median of these numbers. 2 9 4 9 5 1

**1.18** Find the median of these numbers. 20 15 18 11 16 13

## Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

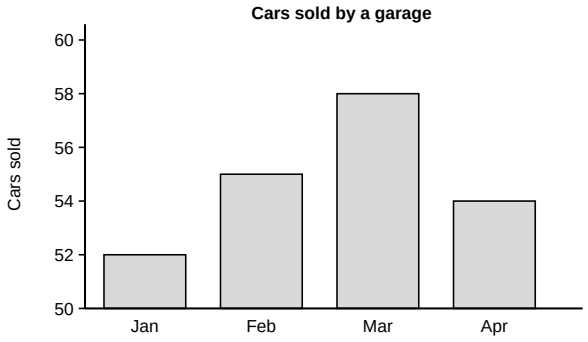
**Worked example: misleading charts**

A bar chart shows the numbers of cars sold. Its vertical axis starts at 50. Give one reason why the bar chart is misleading.

| | | |
|---|---|---|
| **Step 1** | Look at the axis | it starts at 50, not 0 |
| **Step 2** | Answer | small differences look big |

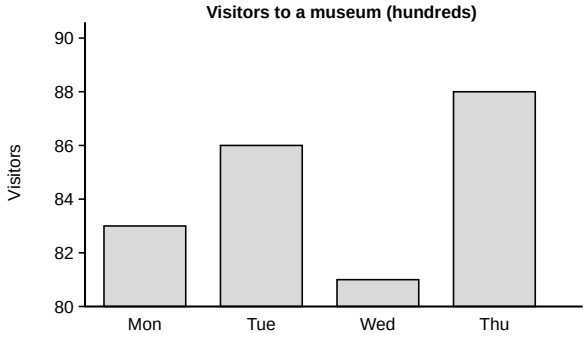
### Practice: misleading charts

**1.19**




The bar chart shows cars sold by a garage.
Give one reason why the bar chart is misleading.

**1.20**




The bar chart shows visitors to a museum (hundreds).
Give one reason why the bar chart is misleading.

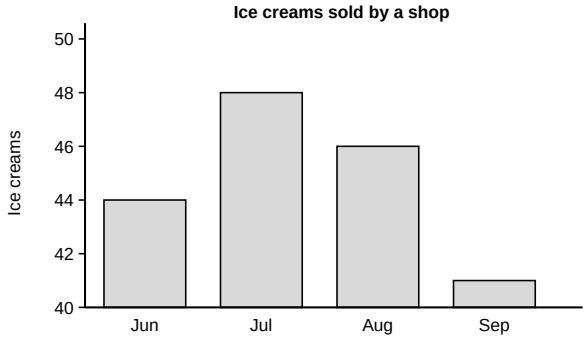
**1.21**




The bar chart shows pizzas sold by a shop.
Give one reason why the bar chart is misleading.

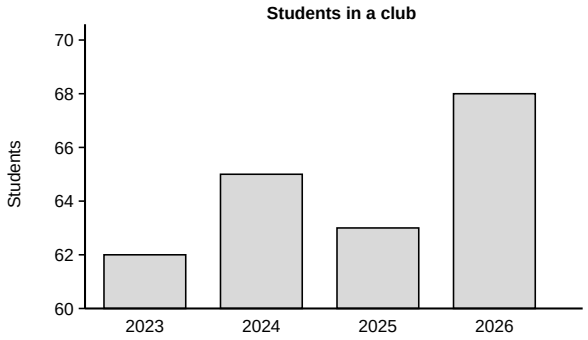
## Practice: misleading charts (continued)

**1.22**



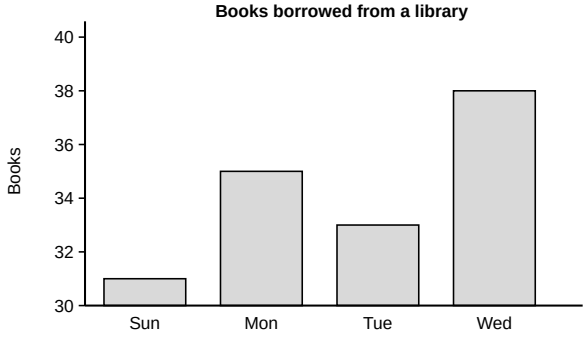

The bar chart shows ice creams sold by a shop.
Give one reason why the bar chart is misleading.

**1.23**




The bar chart shows students in a club.
Give one reason why the bar chart is misleading.

**1.24**




The bar chart shows books borrowed from a library.
Give one reason why the bar chart is misleading.

## Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

### Worked example: questionnaires

Ali uses this question on a questionnaire to find out how much students spend on snacks. Write down two things that are wrong with the question.

How much do you spend on snacks?
0–1 KD ☐ 1–2 KD ☐ 2–3 KD ☐

| | | |
|---|---|---|
| **Step 1** | Is there a time? | No: per day or per week? |
| **Step 2** | Do boxes overlap? | Yes: 1 KD is in two boxes |
| **Step 3** | Is every answer there? | No box for more than 3 KD |

### Practice: questionnaires

**1.25** Huda uses this question on a questionnaire to find out how much time students spend on homework. Write down two things that are wrong with the question.

How many hours of homework do you do?
0–2 ☐ 2–4 ☐ 4–6 ☐

**1.26** Majed uses this question on a questionnaire to find out how often people exercise. Write down two things that are wrong with the question.

How many times do you exercise?
1–3 ☐ 3–5 ☐ 5–7 ☐

**1.27** Layla uses this question on a questionnaire to find out how much people spend on lunch. Write down two things that are wrong with the question.

How much do you spend on lunch?
0–2 KD ☐ 2–4 KD ☐ 4–6 KD ☐

## Practice: questionnaires (continued)

**1.28** Omar uses this question on a questionnaire to find out how much time people spend on their phones. Write down two things that are wrong with the question.

How many hours do you spend on your phone?
0–2 ☐ 2–4 ☐ 4–6 ☐

**1.29** Mona uses this question on a questionnaire to find out how much people spend on clothes. Write down two things that are wrong with the question.

How much do you spend on clothes?
0–10 KD ☐ 10–20 KD ☐ 20–30 KD ☐

**1.30** Fahad uses this question on a questionnaire to find out how many books students read. Write down two things that are wrong with the question.

How many books do you read?
1–2 ☐ 2–3 ☐ 3–4 ☐

## Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

**Worked example: pie charts**

40 people were asked to name their favourite fruit. 12 chose apples. The results are shown in a pie chart. Work out the angle of the sector for apples.

| | | |
|---|---|---|
| **Step 1** | Angle for one: 360° ÷ 40 | $360 \div 40 = 9^\circ$ |
| **Step 2** | × angle for one person | $12 \times 9 = 108^\circ$ |

### Practice: pie charts

**1.31** 30 people were asked to name their favourite colour. 11 chose blue. The results are shown in a pie chart. Work out the angle of the sector for blue.

**1.32** 45 people were asked to name their favourite drink. 16 chose juice. The results are shown in a pie chart. Work out the angle of the sector for juice.

**1.33** 60 people were asked to name their favourite subject. 25 chose maths. The results are shown in a pie chart. Work out the angle of the sector for maths.

**1.34** 40 people were asked to name their favourite fruit. 9 chose bananas. The results are shown in a pie chart. Work out the angle of the sector for bananas.

**1.35** 72 people were asked to name their favourite sport. 20 chose tennis. The results are shown in a pie chart. Work out the angle of the sector for tennis.

**1.36** 20 people were asked to name their favourite pet. 7 chose cats. The results are shown in a pie chart. Work out the angle of the sector for cats.

## Handling Data 4 · Averages and range (Lesson 4.5)

**Worked example: mean and range**

Here are five numbers. 5 9 4 8 4

(a) Work out the mean. (b) Work out the range.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Add all the numbers | $5 + 9 + 4 + 8 + 4 = 30$ |
| | **Step 2** | Divide by how many (5) | $30 \div 5 = 6$ |
| **(b)** | **Step 1** | Biggest − smallest | $9 - 4 = 5$ |

### Practice: mean and range

**2.1** Here are five numbers. 7 3 9 5 6

(a) Work out the mean. (b) Work out the range.

**2.2** Here are five numbers. 12 8 15 10 5

(a) Work out the mean. (b) Work out the range.

**2.3** Here are five numbers. 2 9 4 7 3

(a) Work out the mean. (b) Work out the range.

**2.4** Here are five numbers. 6 11 8 10 5

(a) Work out the mean. (b) Work out the range.

**2.5** Here are five numbers. 14 9 12 7 8

(a) Work out the mean. (b) Work out the range.

**2.6** Here are five numbers. 3 10 5 8 4

(a) Work out the mean. (b) Work out the range.

## Handling Data 4 · Averages and range (Lesson 4.5)

**Worked example: mode and median**

Here are seven numbers. 4 6 6 2 9 6 7

(a) Write down the mode. (b) Find the median.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Most common | $6$ |
| **(b)** | **Step 1** | In order | $2, 4, 6, 6, 6, 7, 9$ |
| | **Step 2** | Middle number | $6$ |

### Practice: mode and median

**2.7** Here are seven numbers. 4 6 4 8 9 4 5

(a) Write down the mode. (b) Find the median.

**2.8** Here are seven numbers. 7 3 3 5 9 3 6

(a) Write down the mode. (b) Find the median.

**2.9** Here are seven numbers. 2 8 5 8 1 8 4

(a) Write down the mode. (b) Find the median.

**2.10** Here are seven numbers. 10 7 12 7 9 11 7

(a) Write down the mode. (b) Find the median.

**2.11** Here are seven numbers. 6 2 9 2 5 3 2

(a) Write down the mode. (b) Find the median.

**2.12** Here are seven numbers. 15 12 15 18 11 15 14

(a) Write down the mode. (b) Find the median.